

Student Number:



Barker College

2012
YEAR 12
TRIAL HSC
EXAMINATION

MATHEMATICS

Staff Involved:

- GIC • JMW
- BJB • DZP
- PJR • ASW
- KJL* • RMH
- LMD*

TUESDAY 31st JULY 2012

130 copies

Multiple Choice Answer Sheet

Choose the best response and fill in the response oval completely

**Start
Here**



- | | | | | |
|-----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |
| 6. | A ☐ | B ☐ | C ☐ | D ☐ |
| 7. | A ☐ | B ☐ | C ☐ | D ☐ |
| 8. | A ☐ | B ☐ | C ☐ | D ☐ |
| 9. | A ☐ | B ☐ | C ☐ | D ☐ |
| 10. | A ☐ | B ☐ | C ☐ | D ☐ |

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General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Write your Barker Student Number on all pages of your answers
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper which may be detached for your use
- Show ALL necessary working in Questions 11-16
- Marks may be deducted for careless or badly arranged working

Total marks - 100

SECTION I Pages 3 – 6

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

SECTION II Pages 7 – 16

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Section I – Multiple Choice
Attempt Questions 1 - 10
All questions are of equal value

Answer each question on the multiple choice answer sheet.

1. The value of $\log_6 2000$ is closest to:

(A) 1.3

(B) 4.2

(C) 5.8

(D) 9.4

2. A circle has the equation $x^2 - 6x + y^2 - 1 = 0$. It has a radius of:

(A) 10

(B) 3

(C) 1

(D) $\sqrt{10}$

3. If $3\sqrt{2} + \sqrt{8} = \sqrt{a}$, the value of a is:

(A) 98

(B) 90

(C) 50

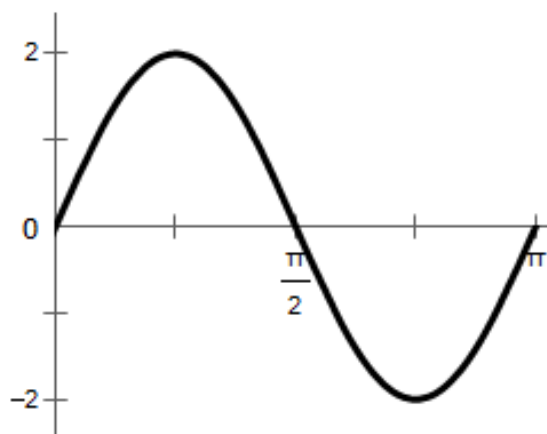
(D) 48

Section I – Multiple Choice continued

4. A function $y = f(x)$ has $f'(3) = 0$ and $f''(3) = -1$. At the point where $x = 3$, $y = f(x)$ is:

- (A) Decreasing and concave up
- (B) Stationary and concave down
- (C) Stationary and concave up
- (D) Stationary with a horizontal point of inflexion

5. The equation of the trigonometric function below could be:



- (A) $f(x) = 2\sin(2x)$
- (B) $f(x) = \frac{1}{2}\sin(2x)$
- (C) $f(x) = \frac{1}{2}\sin\left(\frac{x}{2}\right)$
- (D) $f(x) = 2\sin\left(\frac{x}{2}\right)$

Section I – Multiple Choice continued

6. The range of the function $y = \frac{(\ln x)^2}{-3}$ is given by:

- (A) $y \geq 0$
- (B) $y \leq 0$
- (C) All real y values
- (D) All real y values, $y \neq 0$

7. The value of $\sum_{n=3}^6 \frac{3-n}{2}$ is:

- (A) -3
- (B) $\frac{-3}{2}$
- (C) 0
- (D) 3

8. An expression which is **not** equal to $2^{3x} \times 3^{3x}$ is:

- (A) 6^{3x}
- (B) $2^{3x} \times 9^{\frac{3x}{2}}$
- (C) $\frac{2^{3x} \times 3^{3x-1}}{3}$
- (D) $\frac{2^{3x}}{3^{-3x}}$

Section I – Multiple Choice continued

9. The limiting sum of the series given by $81, 27, 9, 3 \dots$ is

(A) $\frac{243}{2}$

(B) $-\frac{81}{2}$

(C) 120

(D) $\frac{81}{2}$

10. Given the arithmetic sequence $1, 2 + x, \dots$, the tenth term of the sequence will be

(A) $(2 + x)^9$

(B) $(2 + x)^{10}$

(C) $10 + x$

(D) $10 + 9x$

End of Section I

Section II – Extended Response
Attempt Questions 11 - 16
All questions are of equal value

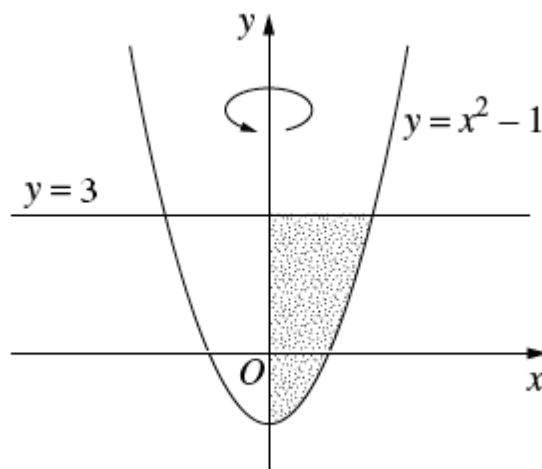
Answer each question in a separate writing booklet. Extra writing booklets are available.

		Marks
Question 11 (15 marks)	[USE A NEW WRITING BOOKLET]	
(a)	Fully factorise $x^3 - 8$	2
(b)	Solve $2\sin x = \sqrt{3}$, for $0 \leq x \leq 2\pi$	2
(c)	Differentiate with respect to x .	
(i)	$2x^3 + \sqrt{x}$	2
(ii)	$\frac{-\cos 2x}{x}$	2
(d)	Solve: $x^2 - 6 > x$	2
(e)	Sketch $y = 2x - 4 $ including any intercepts.	2
(f)	Evaluate $\int_1^2 \left(4t - \frac{3}{t^2} \right) dt$	3

End of Question 11

Question 12 (15 marks) [USE A NEW WRITING BOOKLET]

- (a) Consider the parabola $(y + 7)^2 = 12(x - 3)$.
- (i) Write down the coordinates of the vertex. 1
- (ii) Write down the equation of the directrix. 1
- (iii) Sketch the parabola, showing these features. 1
- (b) Find
- (i) $\int \frac{-x}{2+3x^2} dx$ 2
- (ii) $\int \frac{1}{\sqrt{1+2x}} dx$ 2
- (c) Prove that $\frac{1}{1+\sin \theta} + \frac{1}{1-\sin \theta} = 2\sec^2 \theta$ 2
- (d) In the diagram, the shaded region is bounded by the parabola $y = x^2 - 1$, the y -axis and the line $y = 3$. Find the volume of the solid formed when the shaded region is rotated about the y -axis. 3



Question 12 continues on page 10

Question 12 continued

- (e) A straight line has the equation $y = x + k$.

The perpendicular distance from the point $(2, 7)$ to this straight line is $\frac{1}{\sqrt{2}}$ units.

Find all possible values of k .

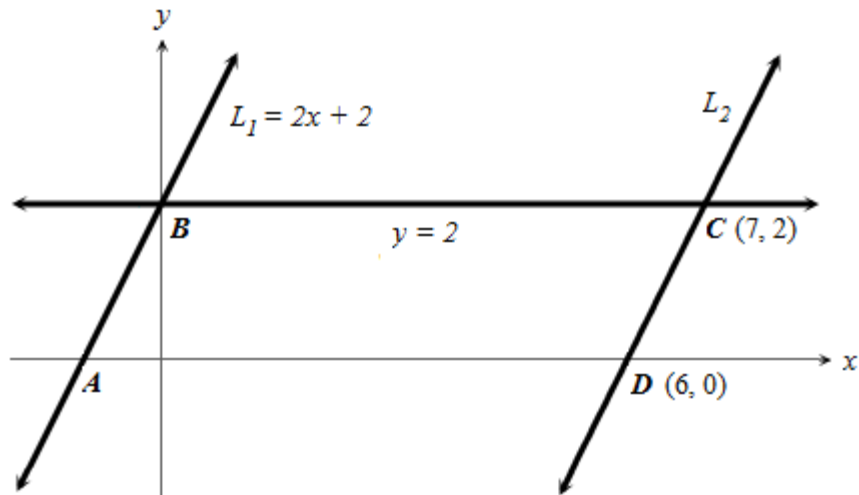
3**End of Question 12**

Question 13 (15 marks) [USE A NEW WRITING BOOKLET]

- (a) In the diagram, two lines L_1 and L_2 intersect with the line $y = 2$.

L_1 has equation $y = 2x + 2$ and meets the x - and y -axes at A and B respectively.

L_2 meets $y = 2$ at the point $C (7, 2)$ and has an intercept at $D (6, 0)$.



- | | | |
|-------|--|---|
| (i) | Find the coordinates of A . | 1 |
| (ii) | Find the equation of L_2 . | 2 |
| (iii) | Find the area of the parallelogram $ABCD$. | 1 |
| | | |
| (b) | Find the equation of the normal to $f(x) = 2 \tan x$, at the point where $x = \frac{\pi}{4}$ | 3 |
| | | |
| (c) | Use Simpson's rule with five function values to approximate $\int_{-2}^2 \sqrt{16 - x^2} dx$. | |
| | Give your answer to 3 significant figures. | 3 |

Question 13 continues on page 12

Question 13 continued

(d) Let $f(x) = x - \ln(x + 1)$

(i) Determine the domain of this function. **1**

(ii) Find the coordinates of any stationary points and determine their nature. **3**

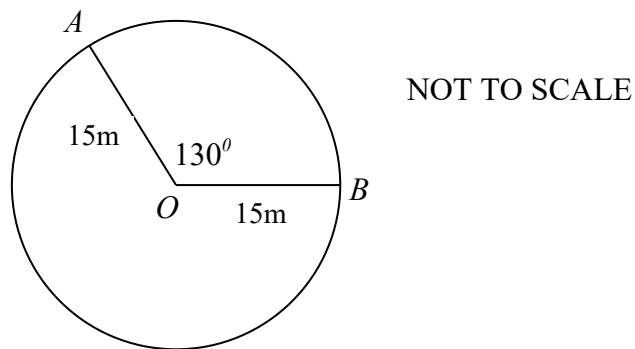
(iii) Hence sketch $f(x) = x - \ln(x + 1)$, including any intercepts. **1**

End of Question 13

Question 14 (15 marks) [USE A NEW WRITING BOOKLET]

(a) Find the exact value of $\cot\left(\frac{2\pi}{3}\right) - \cos\left(\frac{7\pi}{3}\right)$. 2

- (b) Two radii are drawn on a circle of radius 15m, meeting the circle at A and B. They form an angle of 130 degrees at the centre.



- (i) Find the length of the minor arc AB . Leave your answer in exact form. 2
- (ii) Find the area of the minor sector AOB . Leave your answer in exact form. 1

- (c) The quadratic equation $x^2 + mx + k = 0$ has one root that is twice the value of the other.

Prove that $2m^2 = 9k$. 3

- (d) Explain why the function $f(x) = x^3 + 5x$ has no stationary points. 2

Question 14 continues on page 14

Question 14 continued

- (e) The hose from a fire truck is gradually turned off, so that the flow rate of water (V litres) from the hose at time (t seconds) is given by $\frac{dV}{dt} = \frac{t}{2} - 3$.
- (i) At what time was the hose fully turned off? **1**
- (ii) Find V as a function of t if the truck contained 145L of water after 2 seconds. **2**
- (iii) Hence or otherwise, find the total volume of water that is released during the time the hose is turned off. **2**

End of Question 14

Question 15 (15 marks) [USE A NEW WRITING BOOKLET]

- (a) Mr Rundle can see Ms Blowes and Mrs Clarke in the distance. He calculates that Ms Blowes is at a bearing of 030° and Mrs Clarke is at a bearing of 314° .

They are at a distance of 32m and 42m from Mr Rundle respectively.

What is the distance between Ms Blowes and Mrs Clarke?

Give your answer to the nearest metre.

2

- (b) Find the sum of the first 10 terms of this series: $\sqrt{5}, 5, 5\sqrt{5}, 25, \dots$
Leave your answer in exact form.

2

- (c) Over a number of years, Mr Peattie's Rugby Club sells season ticket subscriptions that can be approximated by the exponential growth model,

$$N = Ae^{kt}$$

where N is the number of subscribers and t is the time in years after 1 January 2012.

- (i) It is estimated that at the start of 2012, when $t = 0$, there will be 500 subscribers, and that at the start of 2014, there will be 1200 subscribers. Find A and k , leaving your answers in exact form.

3

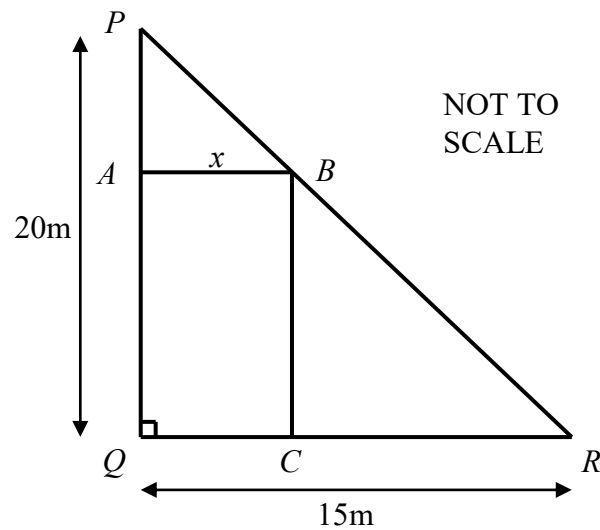
- (ii) According to the model, during which year will the number of subscribers exceed 10 000?

2

Question 15 continues on page 16

Question 15 continued

- (d) In the right-angled $\triangle PQR$, $PQ = 20\text{m}$ and $QR = 15\text{m}$.
 $QABC$ is a rectangle inscribed in $\triangle PQR$, with $AB = x$.

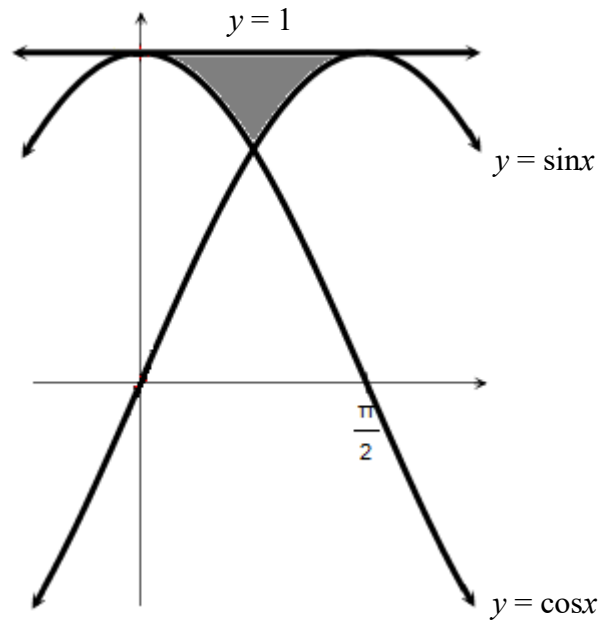


- | | | |
|-------|---|---|
| (i) | Prove that $\triangle PAB \parallel \triangle PQR$. | 2 |
| (ii) | Hence or otherwise, show that $AP = \frac{4x}{3}$ | 1 |
| (iii) | As B moves, the area of rectangle $QABC$ changes.
Find the maximum possible area of the rectangle $QABC$. | 3 |

End of Question 15

Question 16 (15 marks) [USE A NEW WRITING BOOKLET]

- (a) The shaded region in the diagram is bounded by the curves $y = \sin x$, $y = \cos x$, and $y = 1$.



- (i) Find the coordinates of the point of intersection of $y = \sin x$ and $y = \cos x$. **1**
- (ii) Hence or otherwise, find the area of the shaded region. **3**

Question 15 continues on page 18

Question 16 continued

- (b) The line $y = mx$ is a tangent to the curve $y = e^{2x}$.
- (i) Draw a sketch of $y = mx$ and $y = e^{2x}$ on the same set of axes, showing this information. 1
- (ii) Find the value of m . 3
- (c) Ashley has just retired with \$1 000 000 in her bank account. This account attracts interest at a rate of 5% per annum compounded annually.
- She intends to withdraw money in equal amounts at the end of each year, and wants the money in her account to last for exactly 20 years.
- Let A_n be the amount of money in her account after n years.
- (i) Write down an expression for the amount left after two withdrawals. 1
- (ii) Calculate the amount of each withdrawal. 3
- (iii) After her tenth withdrawal, the bank changes its interest rate to 3% p.a. on the balance remaining in her account. Interest is still compounded annually.
- Ashley changes the amount of her withdrawals so that they last the remaining 10 years and are equal in value over these 10 years.
- What will be the new amount of each withdrawal in these last 10 years? 3

End of Question 16
End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Year 12 Mathematics
Trial HSC 2012

Solutions

1. $\log_6 2000 = \frac{\ln 2000}{\ln 6}$

OR $\frac{\log 2000}{\log 6} \approx 4.242...$

(B)

2. $x^2 - 6x + y^2 - 1 = 0$

$x^2 - 6x + 9 + y^2 = 1 + 9$

$(x-3)^2 + y^2 = 10$

circle centre (3,0)

radius = $\sqrt{10}$ (D)

3. $3\sqrt{2} + \sqrt{8}$

$= 3\sqrt{2} + 2\sqrt{2}$

$= 5\sqrt{2} = \sqrt{50}$

(C)

4. When $x=3$,

gradient 0

concave down

(B)

5. Amplitude = 2

Period = π

(A)

6. $y = \frac{(\ln x)^2}{-3}$

numerator always +ve

denominator always -ve

$\therefore y \leq 0$

(B)

7. $\sum_{n=3}^6 \frac{3-n}{2} = 0 - \frac{1}{2} - 1 - \frac{3}{2}$

$= -3$

(A)

8. $2^{3x} \times 3^{3x}$

$= (2^3)^x \times (3^3)^x$

$= 8^x \times 27^x$

(C)

9. $S_{\infty} = \frac{a}{1-r} = \frac{81}{1-\frac{1}{3}}$

$= \frac{243}{2}$

(A)

10. $a=1$ $d=2+x-1=1+x$

$T_{10} = a + (n-1)d$

$= 1 + 9(1+x)$

$= 10 + 9x$

(D)

11a) $x^3 - 8 = (x-2)(x^2 + 2x + 4)$

b) $2 \sin x = \sqrt{3}$

$\sin x = \frac{\sqrt{3}}{2}$

$\therefore x = \frac{\pi}{3}, \frac{2\pi}{3}$

c) i) $y = 2x^3 + \sqrt{x}$

$= 2x^3 + x^{\frac{1}{2}}$

$y' = 6x^2 + \frac{1}{2}x^{-\frac{1}{2}}$

ii) $y = -\frac{\cos 2x}{x}$

$y' = \frac{x(\sin 2x)(2) + (\cos 2x)(1)}{x^2}$

$= \frac{2x \sin 2x + \cos 2x}{x^2}$

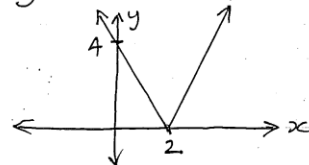
d) $x^2 - 6 > x$

$x^2 - x - 6 > 0$

$(x-3)(x+2) > 0$

$x < -2$ or $x > 3$

e) $y = |2x - 4|$



11f) $\int_1^2 4t - \frac{3}{t^2} dt$

$= \left[\frac{4t^2}{2} - \frac{3t^{-1}}{-1} \right]_1^2$

$= [2t^2 + 3t^{-1}]_1^2$

$= \left(2 \times 2^2 + \frac{3}{2} \right) - (2 + 3)$

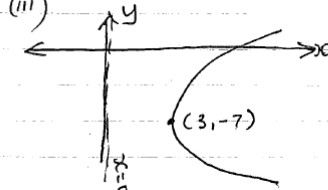
$= \left(8 + \frac{3}{2} \right) - 5 = \frac{9}{2}$

12a) $(y+7)^2 = 12(x-3)$

i) vertex (3, -7)

ii) $a=3$ $\therefore x=0$

iii)



b) i) $\int \frac{-x}{2+3x^2} dx$

$= -\frac{1}{6} \int \frac{6x}{2+3x^2} dx$

$= -\frac{1}{6} \ln(2+3x^2) + C$

ii) $\int \frac{1}{\sqrt{1+2x}} dx$

$= \int (1+2x)^{-\frac{1}{2}} dx$

$= \frac{(1+2x)^{\frac{1}{2}}}{\frac{1}{2} \times 2} + C$

$= \sqrt{1+2x} + C$

c) LHS = $\frac{1-\sin \theta}{1-\sin^2 \theta} + \frac{1+\sin \theta}{1-\sin^2 \theta}$

$= \frac{2}{1-\sin^2 \theta} = \frac{2}{\cos^2 \theta}$

$= 2 \sec^2 \theta = \text{RHS}$

d) $\int_{-1}^3 \pi x^2 dy = \int_{-1}^3 \pi(y+1) dy$

$V = \pi \left[\frac{y^2}{2} + y \right]_{-1}^3$

$= \pi \left[\left(\frac{9}{2} + 3 \right) - \left(\frac{1}{2} - 1 \right) \right]$

$= \pi \left[7\frac{1}{2} + \frac{1}{2} \right] = 8\pi u^3$

e) $y = x + k$

$x - y + k = 0$ (2,7)

$\perp \text{ dist} = \frac{|ax + by + c|}{\sqrt{a^2 + b^2}}$

$\frac{1}{\sqrt{2}} = \frac{|1 \times 2 - 1 \times 7 + k|}{\sqrt{1^2 + 1^2}}$

$\frac{1}{\sqrt{2}} = \frac{|2 - 7 + k|}{\sqrt{2}}$

$\therefore 1 = |-5 + k|$

$k = 4$ or $k = 6$

13a) i) $A(-1, 0)$

ii) $m=2$ $y-0=2(x-6)$

$y=2x-12$

$2x - y - 12 = 0$

iii) $A = bh$

$= 7 \times 2 = 14 u^2$

$$13b) f(x) = 2 \tan x$$

$$f'(x) = 2 \sec^2 x$$

$$f'\left(\frac{\pi}{4}\right) = 2 \sec^2\left(\frac{\pi}{4}\right)$$

$$= \frac{2}{\cos^2\left(\frac{\pi}{4}\right)}$$

$$= \frac{2}{\left(\frac{1}{\sqrt{2}}\right)^2} = 4$$

$\therefore$ grad of normal $= -\frac{1}{4}$
point $\left(\frac{\pi}{4}, 2\right)$

$$y - 2 = -\frac{1}{4}\left(x - \frac{\pi}{4}\right)$$

$$y = -\frac{1}{4}x + 2 + \frac{\pi}{16}$$

$$16y = -4x + 32 + \pi$$

$$4x + 16y - 32 - \pi = 0$$

$$13c) \int_{-2}^2 \sqrt{16-x^2} dx$$

x	-2	-1	0	1	2
y	$\sqrt{12}$	$\sqrt{15}$	4	$\sqrt{15}$	$\sqrt{12}$
weight	1	4	2	4	1

$$h = 1$$

$$A \approx \frac{1}{3}(\sqrt{12} + 4\sqrt{15} + 4 \times 2 + 4\sqrt{15} + \sqrt{12})$$

$$\approx 15.30402333$$

$$\approx 15.3 \text{ (3 sig figs)} u^2$$

$$d) f(x) = x - \ln(x+1)$$

(i) Domain $x > -1$

$$(ii) f'(x) = 1 - \frac{1}{x+1}$$

$f'(x) = 0$ at stat points.

$$1 - \frac{1}{x+1} = 0$$

$$1 = \frac{1}{x+1}$$

$$x+1 = 1$$

$$x = 0, y = 0$$

$\therefore$ stat point at $(0, 0)$

x	-1	$-\frac{1}{2}$	0	1
$\frac{dy}{dx}$	undef	-1	0	$\frac{1}{2}$

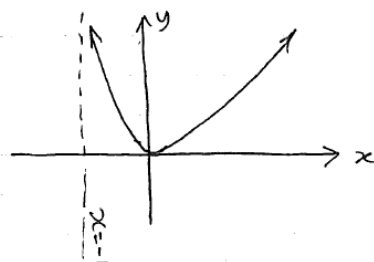
minimum.
 $(0, 0)$

$$\text{OR } f''(x) = -(x+1)^{-2}$$

$$= -(x+1)^{-2}$$

when $x = 0$ $f''(0) = 1$
 $\therefore$ minimum

(iii)



$$14a) \cot\left(\frac{2\pi}{3}\right) - \cos\left(\frac{7\pi}{3}\right)$$

$$= \frac{1}{\tan\left(\frac{2\pi}{3}\right)} - \cos\frac{\pi}{3}$$

$$= -\frac{1}{\sqrt{3}} - \frac{1}{2}$$

$$= \frac{-2\sqrt{3} - 3}{6} = -\frac{\sqrt{3}}{3} - \frac{1}{2}$$

$$b) i) L = r\theta \quad \theta = \frac{130}{180}\pi$$

$$= 15 \times \frac{13\pi}{18} = \frac{13}{18}\pi$$

$$= \frac{65\pi}{6} \text{ m}$$

$$(ii) A = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2} \times 15^2 \times \frac{13}{18}\pi$$

$$= \frac{325}{4}\pi \text{ m}^2$$

$$c) x^2 + mx + k = 0$$

$$\beta = 2\alpha$$

$$(x-\alpha)(x-2\alpha) = x^2 + mx + k$$

$$x^2 - 3\alpha x + 2\alpha^2 = x^2 + mx + k$$

$$-3\alpha x + 2\alpha^2 = mx + k$$

$$\therefore m = -3\alpha$$

$$k = 2\alpha^2$$

$$2m^2 = 2(-3\alpha)^2 = 18\alpha^2$$

$$9k = 9 \times 2\alpha^2 = 18\alpha^2$$

$$\therefore 2m^2 = 9k$$

(OR)

$$\alpha + 2\alpha = -\frac{m}{1}$$

$$3\alpha = -m$$

$$\alpha \times 2\alpha = k$$

$$2\alpha^2 = k \text{ etc}$$

$$d) f(x) = x^3 + 5x$$

$$f'(x) = 3x^2 + 5$$

stat points when $f'(x) = 0$

$$0 = 3x^2 + 5$$

$-5 = 3x^2$ but $3x^2$ is always +ve

$\therefore$ No stationary points.

$$e) \frac{dV}{dt} = \frac{t}{2} - 3$$

(i) turned off when $\frac{dV}{dt} = 0$

$$0 = \frac{t}{2} - 3$$

$$t = 6 \text{ seconds.}$$

$$(ii) V = \frac{t^2}{4} - 3t + C$$

$$\text{when } t = 2 \quad V = 145$$

$$145 = 1 - 6 + C$$

$$150 = C$$

$$\therefore V = \frac{t^2}{4} - 3t + 150$$

$$(iii) \text{ When } t = 0 \quad V = 150 \text{ L}$$

$$\text{When } t = 6 \quad V = \frac{36}{4} - 18 + 150$$

$$= 141 \text{ L}$$

$\therefore$ 9 L of water released

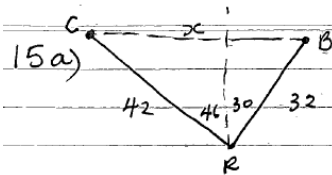
(OR)

$$\int_0^6 \left(\frac{t}{2} - 3\right) dt$$

$$= \left[\frac{t^2}{4} - 3t\right]_0^6$$

$$= \frac{36}{4} - 18 = -9$$

$\therefore$ 9 L of water released.



$$x^2 = 42^2 + 32^2 - 2 \times 42 \times 32 \cos 76$$

$$x^2 = 2137.7139 \dots$$

$$x = 46.235 \dots$$

$$x = 46 \text{ m (nearest m)}$$

b) $a = \sqrt{5}$ $r = \sqrt{5}$

$$S_{10} = \frac{\sqrt{5}((\sqrt{5})^{10} - 1)}{\sqrt{5} - 1}$$

$$= \frac{\sqrt{5}(3125 - 1)}{\sqrt{5} - 1}$$

c) $N = Ae^{kt}$

$$\frac{dN}{dt} = kAe^{kt} = kN$$

ci) When $t=0$ $N=500$

$$A = 500$$

$$\therefore N = 500e^{kt}$$

When $t=2$ $N=1200$

$$1200 = 500e^{2k}$$

$$\frac{12}{5} = e^{2k}$$

$$k = \frac{1}{2} \ln\left(\frac{12}{5}\right)$$

(ii) $N > 10000$

Find the equality:

$$10000 = 500e^{\frac{1}{2} \ln 2 \cdot 4 \times t}$$

$$20 = e^{\frac{1}{2} \ln 2 \cdot 4 \times t}$$

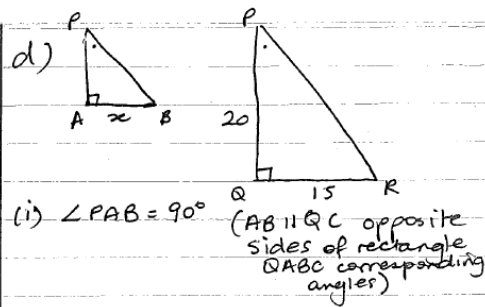
$$\ln 20 = \frac{1}{2} \ln 2 \cdot 4 \times t$$

$$t = \frac{\ln 20}{\frac{1}{2} \ln 2 \cdot 4}$$

$$= 6.843 \dots$$

$\therefore$ During 7th year

$\therefore$ during 2018



(i) $\angle PAB = 90^\circ$ (AB || QR opposite sides of rectangle QABC corresponding angles)

$\angle P$ is common

$\therefore \triangle PAB \sim \triangle PQR$ (equiangular)

(ii) $\frac{AP}{x} = \frac{20}{15}$ (ratios of sides on similar triangles)

$$\therefore AP = \frac{4x}{3}$$

(ii) $A = (20 - \frac{4x}{3})x$

$$= 20x - \frac{4x^2}{3}$$

$$A' = 20 - \frac{8x}{3} \quad A'' = -\frac{8}{3} \therefore \text{maximum}$$

OR When is $A'=0$? For maximum area

$$0 = 20 - \frac{8x}{3}$$

$$20 = \frac{8x}{3}$$

$$60 = 8x$$

$$x = 7.5$$

x	7	7.5	8
A'	$1\frac{1}{3}$	0	$-1\frac{1}{3}$
	/	-	\

maximum

$$\therefore \text{Max Area} = 7.5(20 - \frac{4 \times 7.5}{3})$$

$$= 75 \text{ units}^2$$

16a) $y = \sin x$

$y = \cos x$

$$\frac{\sin x}{\cos x} = \frac{\cos x}{\cos x}$$

$$\tan x = 1$$

$$x = \frac{\pi}{4}$$

(i) $(\frac{\pi}{4}, \frac{1}{\sqrt{2}})$

16a) (ii) Area is symmetrical

$$\therefore \frac{1}{2}A = \int_0^{\frac{\pi}{4}} 1 - \cos x \, dx$$

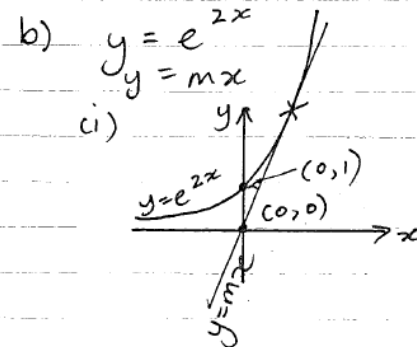
$$= [x - \sin x]_0^{\frac{\pi}{4}}$$

$$= (\frac{\pi}{4} - \sin \frac{\pi}{4}) - (0 - \sin 0)$$

$$= \frac{\pi}{4} - \frac{1}{\sqrt{2}} - 0$$

$$A = 2(\frac{\pi}{4} - \frac{1}{\sqrt{2}})$$

$$= \frac{\pi}{2} - \frac{2}{\sqrt{2}} \text{ units}^2$$



(ii) $e^{2x} = mx$

$$\ln(e^{2x}) = \ln mx$$

$$2x = \ln mx \dots \textcircled{1}$$

$$y' = 2e^{2x}$$

$$m = 2e^{2x} \dots \textcircled{2}$$

$$2x = \ln 2xe^{2x} \textcircled{2} \text{ into } \textcircled{1}$$

$$e^{2x} = 2xe^{2x}$$

$$1 = 2x$$

$$x = \frac{1}{2}$$

$$\therefore m = 2e$$

* There are other *
methods

c) (i) \$1000000

5% p.a. compounded annually

20 years

at end no money left

$$(i) A_1 = 1000000 \times 1.05 - M$$

$$A_2 = (1000000 \times 1.05 - M) \times 1.05 - M \\ = 1000000 \times 1.05^2 - 1.05M - M$$

$$(ii) A_3 = A_2 \times 1.05 - M$$

$$= (1000000 \times 1.05^2 - 1.05M - M) \times 1.05 - M \\ = 1000000 \times 1.05^3 - 1.05^2M - 1.05M - M$$

$$A_{20} = 1000000 \times 1.05^{20} - 1.05^{19}M -$$

$$1.05^{18}M - \dots - M$$

$$= 1000000 \times 1.05^{20} - M(1 + 1.05 +$$

$$1.05^2 + \dots + 1.05^{19})$$

$$= 1000000 \times 1.05^{20} - M \left(\frac{1(1.05^{20} - 1)}{0.05} \right)$$

After 20 years $A_{20} = 0$

$$0 = 1000000 \times 1.05^{20} - M \left(\frac{1.05^{20} - 1}{0.05} \right)$$

$$M = \frac{1000000 \times 1.05^{20}}{\left(\frac{1.05^{20} - 1}{0.05} \right)}$$

$$= \$80242.58719$$

$$(iii) A_{10} = 1000000 \times 1.05^{10} \\ - 80242.59 \left(\frac{1.05^{10} - 1}{0.05} \right) \\ = \$619611.95$$

$$A_1 = 619611.95 \times 1.03 - M$$

$$A_2 = A_1 \times 1.03 - M$$

$$= (619611.95 \times 1.03 - M) \times 1.03 - M$$

$$A_3 = 619611.95 \times 1.03^2 - M \times 1.03 - M$$

$$A_3 = 619611.95 \times 1.03^3 - 1.03^2M - 1.03M -$$

$$A_{10} = 619611.95 \times 1.03^{10} - M(1 + 1.03 + \dots + 1.03^9)$$

$$= 619611.95 \times 1.03^{10} - M \left(\frac{1(1.03^{10} - 1)}{0.03} \right) \\ \text{But } A_0 = 0$$

$$0 = 619611.95 \times 1.03^{10} - M \left(\frac{1.03^{10} - 1}{0.03} \right)$$

$$M = \frac{619611.95 \times 1.03^{10}}{\left(\frac{1.03^{10} - 1}{0.03} \right)}$$

$$= \$72637.42$$